

# Recognition-Independent Handwritten Text Alignment Using Lightweight Recurrent Neural Network

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## ABSTRACT

Legibility refers to the ease with which handwritten content can be read and understood accurately. However, existing approaches to handwriting beautification either rely on the result of handwriting recognition and accumulate errors from the recognition system or do not address the alignment problem and are difficult to generalize to other languages. This paper presents a novel approach to improve handwriting legibility by straightening the written content. It utilizes a recurrent neural network that operates without the need for recognition, supports connected writing, and accommodates various writing styles. The results obtained with this method demonstrate significant improvements in handwriting alignment. Moreover, a single neural network model can effectively cater to multiple languages within the same writing system.

## CCS CONCEPTS

• **Human-centered computing** → **Text input**; • **Applied computing** → *Graphics recognition and interpretation*.

## KEYWORDS

Handwriting beautification, recurrent neural networks

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## 1 INTRODUCTION

In recent years researchers have managed to achieve high-quality results in handwriting recognition, verification, and other handwriting-related tasks. There is a lot of conducted research devoted to handwritten trace improvement, which can be used to make handwriting more legible and better-looking for the human eye. It is especially

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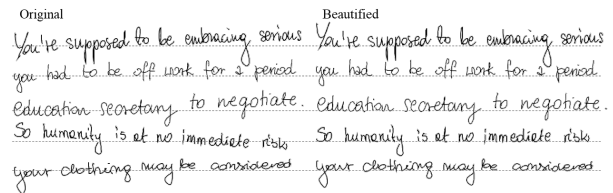


Figure 1: Examples of handwritten text alignment.

important in medicine and education, where poor writing can lead to prescription errors and problems with performance. Handwriting legibility is closely associated with various stylistic attributes of writing, including letter shape, size, slope, and the spacing between letters, words, and lines. In a paper [Lu et al. 2012] the style transfer method was proposed to generate and beautify handwriting and line drawings. Token Means approach [Zitnick 2013] is based on averaging multiple matching tokens (a sequence of stroke samples). However, it does not address the alignment problem, which has a strong visual impact while talking about beautification. The approach proposed in [Simard et al. 2005] includes an ink extremum classifier for tracking baseline and midline. Such an approach has no need for previous segmentation and recognition, however, it is difficult to generalize to other languages, since it requires a lot of time and effort to collect and label the dataset.

In this paper, we introduce a novel approach aimed at enhancing the legibility of handwriting by straightening the written content in order to raise user satisfaction and make the text more legible, neat, and good-looking (Figure 1). Our method utilizes a bidirectional Gated Recurrent Unit (GRU), obviating the need for character segmentation.

## 2 APPROACH

Our goal in this work is to develop a method for handwritten text alignment which operates without the need for prior character recognition or segmentation, and supports connected writing, accommodating various writing styles. We assume that the neural network (NN) model should determine the relations in data all by itself, and perform text straightening only on the basis of the data, without knowing any additional information about character segmentation or label.

Based on the above definition, we propose to use a two-layer bidirectional GRU model with a hidden state size equal to 40 to

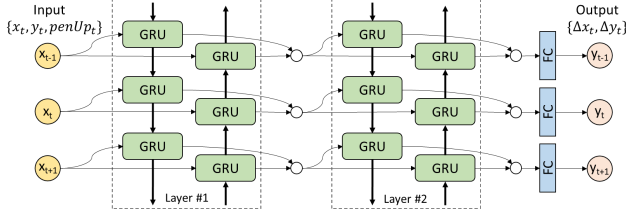


Figure 2: Architecture of the proposed neural network.

build an efficient handwritten text correction model (Figure 2). Having data passed through the GRU layer, we take the GRU outputs and pass them through the additional linear layer to project an 80-dimensional feature vector to the output 2-dimensional space for each point. The total number of trainable parameters of the denoising model is only about 40k. As a model input, we propose to use  $\{x_i, y_i, p_i\}_{i=0}^T$ , where  $x_i, y_i$  represent the trajectory of the pen movement ( $x$  and  $y$  coordinates) and  $p_i$  represents the air-state of the pen at a specific time stamp  $i$ . We define model output as follows:  $\{\Delta x_i, \Delta y_i\}_{i=0}^T$  for  $i$ -th point of stroke size  $T$ , which shows how much each point needs to be offset to compensate for the shift.

Having this done, we need to generate training pairs of data. We pass the input data through the structural text straightening system to get the reference sequence for training and verification. In such a way, we prepared a training set of about 530k sample pairs written in English. The dataset includes samples with labels 2 to 64 characters long, with an average label length equal to 9.59. To make the intermediate data representation invariant to scale movement and distortion, the strokes are centered and scaled: for each sample, its  $x_{min}$  and  $y_{min}$  are set to 0, and all the data are scaled with a factor of specified standard height (set to 100.0).

Once we have the model trained, we can use it for handwritten text beautification. After passing input data through the network, we have a specific shift for each point we need to apply to straighten the input. To avoid stroke distortion, we propose to move each point by the average stroke shift  $\{\Delta x_{mean}^i, \Delta y_{mean}^i\}_{i=0}^M$ , where  $M$  is a number of strokes.

### 3 EVALUATION AND DISCUSSION

The evaluation of the performance of our method is two-fold. Firstly, we use Euclidean distance to measure time series similarity. We used 540 handwritten samples (original), generated straightened samples (reference), and outputs of the neural network (aligned). All samples were preliminarily centered and normalized by their heights. Two sets of distances were calculated from the original to the reference handwritten sample, and from the aligned to the reference. The results are presented in Figure 3a. The horizontal axis corresponds to the sample number (samples were sorted by Original-Reference distance in ascending order).

Secondly, we conducted a user study with the main objective of assessing the visual appeal and readability of aligned text. During the survey, each user was presented with 106 examples of single-line text in English to compare and evaluate. For each of the selected examples, we applied the proposed algorithm, generating modified versions of the handwriting. Participants were presented with both

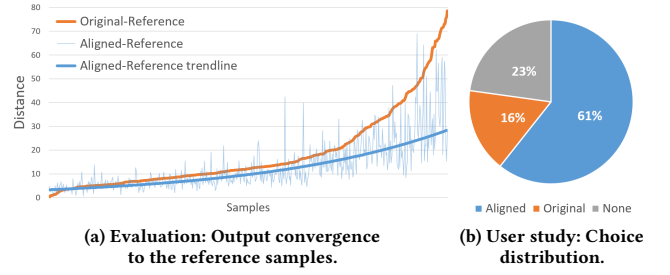


Figure 3: Experiments.

the original and modified versions and were asked to select which one looked better, or *None* option if they could not choose (no preference). The randomization prevented participants from knowing which version was the original and which was the beautified one.

The survey involved the participation of 18 users. The summarized results are shown in Figure 3b. 61% of aligned versions were preferred by the participants over the original handwriting, whereas 16% of the original was chosen. In particular, only 6.6% of all original versions were preferred by more than 60% of users. On the other hand, 57.5% of all aligned versions were chosen by more than 60% of participants, confirming that the results of handwritten alignment provided by our method look better for the majority of participants. The large value of "None" option selections can be explained by the part of initially well-written samples, so the resulting output of our algorithm doesn't have a big difference. Therefore both variants look similar for users, and neither could be preferred.

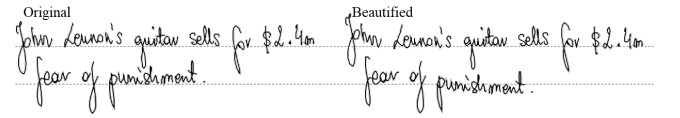


Figure 4: Examples of failure cases.

The proposed approach demonstrates notable enhancement particularly when dealing with highly uneven texts. In cases where the original text is already relatively uniform, the method maintains the writing style without inducing substantial alterations. Notably, the average error per symbol slightly surpasses one pixel, a discrepancy nearly imperceptible to the human eye.

As further research, we plan to explore the unification of not only letter positions but also letter size and shape, enhance the quality of the NN model, explore alternative architectures, and investigate the feasibility of using this approach as a preprocessing step before recognition.

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